

An integral 5-adic point on the five-row comparison variety

Exact computational certificate; independent review requested

30 September 2026

Proposition 1. *The fifteen centered five-row, four-column comparison moment equations have a solution with all coordinates in $\mathbb{Z}_5$.*

Proof. For nonempty $S \subseteq \{1, 2, 3, 4\}$ use the integer polynomials

$$F_S(x) = 3 \sum_{i=1}^5 \prod_{j \in S} x_{j,i} - b_S, \quad b_S = \begin{cases} 5, & |S| = 2, \\ 3, & |S| = 4, \\ 0, & |S| \text{ odd.} \end{cases}$$

Take the initial four columns

$$(22, 15, 33, 31, 24), \quad (97, 65, 58, 31, 124), \quad (14, 6, 88, 0, 17), \quad (62, 75, 113, 106, 19).$$

In subset order by size and then lexicographically their residuals are

$$(375, 1125, 375, 1125, 26875, 11125, 29875, 26875, 69250, 34750, \\ 764250, 1740375, 1085250, 2190750, 66855375).$$

All residuals are divisible by 125. Flattening the variables column by column with indices $0, \dots, 19$, choose the fifteen Jacobian columns

$$0, 1, 2, 3, 5, 6, 7, 10, 11, 12, 15, 16, 17, 4, 8.$$

The resulting square matrix J has determinant

$$-254169067184538400674529220966400,$$

of 5-adic valuation two, and has rank thirteen modulo five. Hence every cofactor is divisible by five, so all entries of J^{-1} lie in $5^{-1}\mathbb{Z}_5$.

Fixing the other five variables, the map $y \mapsto y - J^{-1}F(y)$ preserves the ball $x + 25\mathbb{Z}_5^{15}$ and is a strict contraction there. Indeed its initial displacement has valuation at least two, and the change of the polynomial Jacobian throughout the ball has entries divisible by 25. The contraction principle supplies an exact zero of all the moment polynomials.

The integer certificate is stored in `general_q5_hensel_certificate.json`; the script `lift_general_q5.py` reproduces the residual, determinant, and inverse valuation checks by exact rational arithmetic. $\square$

This is a statement over $\mathbb{Q}_5$, with no assertion of a rational or real monotone point. Together with the separate 2-adic and 3-adic certificates, it rules out local insolubility at the three smallest primes as an obstruction to five-row local models.